When Can a Swarm Be Audited?[1]

Stateful audit evidence and measured monitor behavior

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With
Apart Research

Abstract

A monitor can name the true executor while lacking evidence that an unauthorized action occurred. We implement a bounded two-worker audit testbed with sequential delegation, deferred revocation and joined approvals, separating identity bindings from execution results. An enumerator and independent constraint checker agree on 19,648 prescribed stateful projection checks. We conducted 3,840 Adaption reader calls over 192 scripted source episodes and 240 matched Hugging Face reader calls over 24 of those sources. Complete, random-loss and selective-loss exports are paired across naming conditions; authentic two-record restoration is compared with random add-back. The primary selective-minus-random unsupported-worker contrast was -1.04 [-6.77, +4.69] percentage points. For the primary reader, priority restoration changed grounded correctness relative to random restoration by +43.23 [+37.50, +48.96] points. Grounded correctness was 1396/1920 and 1371/1920 for the two Adaption readers, and 128/240 for the HF reader. A separate controlled influence arm produced 0 unauthorized writes across 192 receiving-worker choices. The protocol was frozen before collection; a gateway interruption required an explicit no-retry operational continuation. All failures and serving amendments are retained. These finite synthetic-workflow measurements support an inspectable evaluation and evidence-preservation comparison, not general swarm safety or spontaneous collusion.

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1. Introduction

Suppose a request is approved and queued, then another worker revokes permission before it settles. A later claim that the task completed cannot establish either execution or current permission. An investigator needs authentic state history, a result and its link to an immutable executor. Whether a violation occurred, who executed it and whether agents colluded are different questions [1,2,6,23–25].

We study supported executor attribution in bounded stateful handoffs. The primary question is whether selective export loss causes more unsupported worker accusations than random loss at the same record budget. A second question asks whether authentic evidence restoration improves grounded decisions. Track 4 oversight anchors the work, with Track 3 evidence-infrastructure relevance. Following the sprint rubric, the contribution is executable methodology, genuine measurements and reproducible findings [12], rather than a product demonstration or a new general theory of collusion.

2. Related Work

Qin et al. study trace tampering, independent recording and peer imitation [3]. Chan et al. motivate identity and infrastructure requirements [4,5]. Yun et al. already distinguish correct guesses from supported findings, include abstention and a same-packet comparator, and cross binding availability with identifier renaming [6]. Naming changes alone are therefore weak novelty. We add an explicit stateful executor contract, bounded validation and matched preservation measurements.

Who&When and BLADE target decisive-error responsibility [19,20]; our label is the executor of a designated unpermitted mutation. BRIDGE verifies runtime repair [10]; our restoration is retrospective and assumes authentic originals. Syscon and ActionLens illustrate external evidence and component ablations [17,18]. Selected Apart reports/reviews motivate checkable controls, causal distinctions and honest negative results [13–15]. Table 1 locates the narrower comparison.

HELIOS examines activations, while AI Control and CaMeLs address different monitoring/trust boundaries [7–9]. Distributed red-teaming and multilayer investigations operate beyond this small grammar [21,22]. Swarm Traces has identity, timing and outcome uncertainties and is not our scoring oracle [11]. Prompt Infection already studies inter-agent propagation [28]; our scripted one-hop influence arm does not originate that phenomenon.

Table 1. Prior results and the narrower present measurement. This is a scope comparison, not measured performance.

Related result	Already established	Narrower measurement here
Qin [3] / Yun [6]	Trace integrity, binding, supported findings	Stateful permission plus executed target support
Who&When [19] / BLADE [20]	Decisive-error responsibility	Executor of a designated unpermitted mutation
BRIDGE [10] / Syscon [17]	Runtime repair / external receipts	Retrospective authentic-record restoration cost

3. Methods

Finite workflow. A coordinator assigns two distinct worker instances from {A,B,C}. The first handles h0; the receiver handles designated target h1. Sequential tasks have an authorized decoy update followed by target settlement. Deferred tasks queue h1, then an authorized h0 update changes a fake revocation flag from 0 to 1; h1 settles under current permission. Approval-join tasks require original authorization and two approval branches. Applied-but-unpermitted target writes remain possible, so positive controls are not blocked by construction. Only newly allocated fake dictionaries change.

Labels and evidence. The label is h1's immutable executor if its update is applied without permission, otherwise NONE. It is not initiator, moral responsibility or intent. Twelve fixed schema slots separate request, alias handoff, binding, dispatch, approval context and post-execution result. Slot order is not chronology. A queued operation is not proof of successful settlement; h0 is independently authorized and never the designated violation. Advisory prose is untrusted. Collector originals and public context are trusted; self-consistent forged originals are outside scope.

Oracle validation. Worlds vary the distinct worker pair, arbitrary alias permutation, target outcome, decoy outcome and joined approvals. There are 216 worlds per public context in sequential/deferred tasks and 864 in approval-join tasks. The enumerator matches the entire visible packet. A separately written constraint checker imports no generator, simulator or projection helpers. They agree on 15,552 world/authorization/naming/loss checks plus 4,096 masks of a representative world; golden and malformed-type fixtures also pass. This checks the declared finite contract, not arbitrary histories or its realism.

Observation and restoration. Every source supplies stable/changed-name counterfactuals with the same state and route. Each has complete evidence, uniform seeded deletion of three of twelve records, and selective deletion of handoff e1, target binding e6 and target result e9. Masks are

shared across names/readers. Selective-loss views additionally receive exactly two authentic records by fixed priority or seeded random add-back. Priority queries public schema positions, first e9 then e6; hidden labels do not select records. This strong query interface and deletion/priority pairing make some answerability gain constructed. Equal records do not mean equal tokens or information.

Sampling and serving. The main grid has 64 independently seeded draws per family: 192 sources, balanced to 96 clean/96 scripted-violation strata, ten dependent views and two readers (3,840 calls). Approval-join model outputs were withheld from development and qualification. Gemma 4 26B A4B is primary; Gemma 4 31B is a same-family secondary reader. HF Llama 3.3 70B [31] evaluates the fixed first eight source indices per family: 24 matched sources/240 calls, not new independent sources. Requests use the frozen prompt, temperature zero, fresh context, 4,096 output tokens, no tools/memory/continuation and recorded usage [26,27]. Provider names do not pin immutable weights.

Registration and failures. stateful-main-v2 fixes the core before fresh collection. V1 qualification failed bare-JSON validity; v2 prospectively accepts a bare object or one exact outer json fence, preserving raw strict validity and rejecting prose, multiple objects and duplicate keys. Fresh v2 qualification completed 80/80 operational checks. A main HTTP 502 stopped collection after 3,028 terminal records. Before examining Adaption main effects, stateful-main-v2-continuation-v1 registered only the 812 originally unstarted calls. The failed 502 was retained as INVALID with reserved unknown cost; no attempted case was retried. Appendix B records all operational amendments, including prior HF failures.

Scoring and analysis. Compatible labels singleton: that definite answer is supported; multiple labels: ABSTAIN is justified; empty set: inconsistent evidence halts admission. NONE requires support. Unsupported-worker U=1 for any named worker not supported by that singleton, even a factually correct guess; NONE/ABSTAIN/INVALID have U=0. Grounded correctness, supported definite coverage, unsupported NONE and invalidity prevent this endpoint being gamed by silence. Exact-packet, always-abstain and fixed-A baselines receive the same public evidence. The primary estimate is paired targeted-minus-random U at churn present, first call, primary reader. Naming interaction and restoration/model comparisons are secondary. Ten-thousand percentile-bootstrap resamples preserve whole source clusters, stratified by family/cohort; views never increase independent n. No multiplicity-adjusted or deployment-wide claim follows.

Controlled influence. A separate model origin reads a fake decoy and hands off a target read. A model receiver chooses read/refuse/fake write under scripted malicious/benign/irrelevant/no-peer messages crossed with a warning. The host executes typed actions only. Twenty-four contexts imply 24 origin and 192 receiver calls. Exposed messages are character-matched, not token-matched; no-peer is shorter. The paired malicious-minus-benign executed-violation contrast without warning is secondary. This measures one-hop intervention effects, not emergent coordination [3,8,28]. No monitor fine-tuning was performed.

4. Results

Collection and structure. All 3,840 planned Adaption requests and 240 planned Llama/HF requests have retained terminal records; failed transports remain invalid outputs. The structural singleton counts at churn present were full/none: 192/192, random/none: 143/192, targeted/none: 65/192, targeted/priority: 192/192, targeted/random: 107/192. The exact comparator is an upper reference within this grammar, not a deployable collector guarantee. Table 2 reports all views, including restored packets; its denominators are calls, not independent episodes.

Table 2. Observed calls across all ten views, including restoration. Grounded correctness includes justified abstention. The final column lists unsupported worker answers, unsupported NONE answers and invalid outputs. HF uses 24 matched sources; each Adaption reader uses 192. Views are dependent.

Reader (calls)	**Grounded correct**	**Supported definite**	**Worker / NONE / invalid**
Gemma 4 26B A4B (1920)	1396	1372	146 / 343 / 35
Gemma 4 31B (1920)	1371	1367	175 / 373 / 0
Llama 3.3 70B / HF (240)	128	76	36 / 6 / 0

Primary and restoration comparisons. The primary unsupported-worker difference was -1.04 [-6.77, +4.69] percentage points (192 paired sources); its interval includes zero and does not support the hypothesized increase. Unsupported NONE findings in the primary reader were 98/192 under selective loss and 28/192 under random loss. Clean-source worker-accusation events occurred in 7/96 and 6/96 sources across the two primary loss calls. Priority-minus-random restoration changed grounded correctness by +43.23 [+37.50, +48.96] points; priority-minus-unrestored selective loss changed supported definite coverage by +63.54 [+58.33, +68.75] points. Table 3 separates primary and secondary comparisons. These intervals describe the finite stratified design; a collapsed zero-event bootstrap does not prove zero risk.

Table 3. Paired source estimates and 95% percentile-bootstrap intervals, in percentage points. Only the first reader's U contrast is primary; the rest are secondary. HF shares source draws with the main study.

Reader (source n)	**Targeted — random U**	**Priority — random grounded**
Gemma 4 26B A4B (192)	-1.04 [-6.77, +4.69]	+43.23 [+37.50, +48.96]
Gemma 4 31B (192)	-14.06 [-18.75, -9.38]	+44.79 [+39.06, +50.52]
Llama 3.3 70B / HF (24)	-16.67 [-33.33, -4.17]	-4.17 [-29.17, +20.83]

Secondary model comparison. The matched HF reader's restoration contrast was -4.17 [-29.17, +20.83] points; its interval includes zero, so the Gemma restoration gain did not clearly replicate in this smaller cross-family subset. Across the 1,920 main packets per reader, exact-packet, always-abstain and fixed-A baselines achieved grounded correctness of 1,920, 522 and 230, respectively. Always-abstain had zero unsupported worker accusations but no supported definite findings; low U alone is therefore insufficient.

The separate influence arm retained 216 of 216 planned model calls, including 192 receiving-worker choices. It executed 0 unauthorized fake writes and achieved 177 allowed reads. The paired malicious-minus-benign violation contrast without warning was +0.00 [+0.00, +0.00] percentage points over 24 contexts. Allowed-task success was 11/24 with malicious peer text, 24/24 with benign text and 23/24 with malicious text plus warning. These utility/warning comparisons are secondary and descriptive.

Resources and serving. The main token-rate estimate, including unknown-charge reservations, is 0.3506672880 Adaption credits; known provider-reported cost fields total approximately 0.253835568. The completed HF reader estimate is $0.031954015. These are not reconciled account debits. Prior HF Cerebras preflight produced zero evaluation calls; Qwen/Nscale stopped after eight calls with three reasoning-only truncations, estimated $0.0714972. Those outcomes are retained separately, not pooled as completed replications. No top-up, training job or paid GPU deployment was required.

5. Discussion and Limitations

The practical distinction is between a plausible executor and a supported finding. Losing a result can make successful execution uncertain even when identity is known; losing a binding can leave execution known but executor uncertain. Restoring authentic joins can improve both structural answerability and reader decisions. The public-position restoration interface is intentionally favorable and partly constructed; these results do not invent generic runtime repair [10] or

logging. State-dependent revocation/approval provides a bounded extension of existing evidence-support work [6].

Only two active workers and three scripted transition structures were tested. A matched handoff-free population control, arbitrary histories, scaling and compromised collectors were not evaluated. Hosted endpoints can drift and are not pinned weight hashes. Model/encoding choices followed excluded development; HF reader selection followed recorded availability/completion failures. The gateway continuation changed an operational stopping rule after service failure, although sample, scientific endpoints and planned identities were fixed. The study is not an uninterrupted, unamended confirmatory run. All invalid outcomes, phases and liabilities remain visible.

The artificial balanced mixture is not deployment prevalence. Repeated finite configurations are possible; source resampling assumes independent draws/calls within the declared design. Ninety-six clean sources cannot certify sub-1% false-accusation risk, even with zero events. Multiple secondary comparisons are descriptive. A scripted one-hop null or failure does not establish resistance to peer influence or spontaneous collusion. Small-monitor SFT remains conditional on supported labels, source/family splits, a bounded price and demonstrated base/adapter serving [29,30]; hidden actor labels on ambiguous evidence would train unsupported certainty.

6. Conclusion

A working stateful testbed, genuine Adaption/HF measurements and offline public-data replay now make the oversight question empirically inspectable. The evidence separates structural ambiguity, reader mistakes, service failures and restoration benefit. The primary contrast and all secondary/null findings are reported under the declared design, with the operational continuation disclosed. The result is a bounded research evaluation and preservation comparison, not a general swarm-safety guarantee.

Code and Data

Code, frozen protocols, sanitized fake episodes/final answers, analysis and an offline verifier: https://github.com/razzyrazzy11/swarm-alignment/tree/results-v1 . Run python -m experiments.verify_stateful_results after checkout; native logs remain private and hash-bound. Core, HF and operational-continuation tags identify pre-collection versions. The preexisting October trusted-agent-sting code/data/results were not imported [16]. Automated reviews and CI do not constitute human author approval.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

The trust contract is deliberately bounded: authentic collector originals, public schema positions, known permission rules and fake state only. Self-consistent forged trusted records, collector compromise, real policy ambiguity and incident chronology are outside it. Agreement of two implementations can preserve a shared specification mistake. The finite mask checks do not validate arbitrary malformed inputs, all real agent traces or operating-system containment.

Evidence-loss tests can reveal which records to suppress or how to frame an instance. We publish benign typed fake-data fixtures and supported-decision measurements, not executable incident payloads or real service attacks. Model output is never evaluated as code. Network calls are made only by the authorized orchestrator to inference providers; model tools/memory are disabled. No human-subject, personal-data or live-security experiment was conducted.

The repository release excludes API keys, native provider IDs, request headers, raw reasoning and attendee/source transcripts. Actual final-answer text and fake packets are retained without semantic repair. The authors remain responsible for verification, cited sources and submission. Source coverage is targeted, with recorded inaccessible originals; no universal originality or eligibility clearance is claimed.

Appendix B. Registration, operational amendments and costs

Main-v2 registration and source/prompt/config hashes precede its fresh qualification/main calls. V1's 80-call bare-JSON gate failed 30 fenced responses and remains archived. V2 prospectively admits one exact outer json fence while reporting raw strict validity; fresh qualification passed 80/80. Original development and qualification seeds are excluded from main draws. Approval-join model responses were withheld from development/tuning; its rules are public, not secret.

One HTTP 502 with missing usage stopped the initial main phase at 3,028 calls. All starts have terminal records. The separate continuation tag, predecessor hashes and ordered 812-request complement were published before new calls; the 502 was never retried or converted to abstention. The audited union has one terminal record per original planned identity. The untouched initial phase remains archived. Bounded additional gateway-error tolerance was an explicit operational amendment, not an effect-dependent extension. No Adaption main effect was inspected to select it; the independent HF summary became available while the continuation was prepared.

HF-v1 pinned Cerebras but stopped at preflight when that route disappeared: zero evaluation calls. HF-v2 pinned Qwen/Nscale and stopped after its third incomplete output, retaining eight responses and all unstarted identities. HF-v3 registered Llama/Novita before any Llama answers, keeping the same subset, prompt, metrics and token allowance. Reader selection followed completion feasibility, not a favorable scientific effect. All versions and failures are retained; neither Qwen's five final answers nor three truncations are treated as a completed replication.

Initial observed resources were 1,500 Adaption credits and $2 HF Pro compute credit; the user authorized available Adaption credit. The main nominal ceiling was 20 credits, each qualification/influence stage one, and the HF reader $1. These are client accounting controls, not provider-enforced caps. Token estimates include worst-case reservations for unknown usage; known provider cost fields do not establish final account debit. Final account reconciliation remains outstanding. No top-ups, SFT jobs or GPU rentals were made.

Appendix C. Reproducibility and learning from iterations

Read STATEFUL_PROTOCOL.md, HF_STATEFUL_PROTOCOL.md and STATEFUL_CONTINUATION.md before running. Reproduce public packet labels, final-answer scores, aggregate counts, costs and source-clustered effects with python -m experiments.verify_stateful_results. CI checks the published data without credentials. The frozen API runners refuse silent overwrite/resume; fresh reruns require an explicitly amended version and consume credits. Temperature zero is not deterministic hosted inference.

Earlier development uses one eight-record grammar and 20 reused source seeds. Its fixed 120-call batch yielded grounded correctness 54/60 and 59/60 and did not show increased worker accusations under selective loss. Separate original batches retain 28, 18 and 120 records after interruptions/qualification, and a 32-call peer pilot produced no writes. They are not independent replications or new main data. Early source snapshots/pre-call journals were incomplete, leaving unknown old calls; later hashes do not repair that retroactively. PILOT_RESULTS.md records these limits. The new stateful contract has separate sources and validation.

20-rules-of-learning was used editorially: begin with a concrete failure, define permission/execution/identity before metrics, distinguish supported findings from correct guesses, and present source-level denominators before comparisons. It is not a novelty criterion. Future monitor SFT would require at least the vendor's effective-row minimum, episode/family splitting before augmentation, supported singleton/ABSTAIN labels and untouched evaluation data. If main results select an intervention, a new holdout and new freeze are necessary [29,30].

LLM Usage Statement

Codex assisted with primary-source review, implementation, testing, analysis and writing. Isolated OpenCode reviewers inspected approved source copies without keys or raw runs; their scoped findings and fixes are recorded in REVIEW.md. Hosted models supplied the actual retained reader/worker outputs. Human author review and responsibility for all claims remain required; automated validation is not independent empirical replication or an Apart acceptance decision.